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Phase transformation of GeO₂ glass to nanocrystals under ambient condition

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ABSTRACT: Theoretically, the accomplishment of phase transformation requires sufficient energy to overcome the barriers of structure rearrangements. Transition of an amorphous to crystalline structure is implemented traditionally by heating at high temperatures. However, phase transformation under ambient condition without involving external energy has not been reported. Here, we demonstrate that phase transformation of GeO₂ glass to nanocrystals can be triggered at ambient condition when subjected to aqueous environments. In this case, continuous chemical reactions between amorphous GeO₂ and water are responsible for the amorphous to crystalline transition. Dynamic evolution process is monitored by using in-situ liquid-cell

transmission electron microscope, clearly revealing this phase transformation. It is the hydrolysis of amorphous GeO_2 that leads to the formation of clusters with a size of ~ 0.4 nm, followed by the development of dense liquid clusters, which subsequently aggregate to facilitate the nucleation and growth of GeO_2 nanocrystals. Our finding breaks the traditional understanding of phase transformation and will bring about a significant revolution and contribution to the classical glass crystallization theories.

KEYWORDS: GeO_2 , phase transformation, crystallization, nanocrystal, in-situ TEM

Phase transformation is a process that involves microstructural transition of materials when subject to external environments and usually accompanied by a variation in energy.¹⁻³ Whenever a phase transformation occurs, sufficient energy is required to overcome the barriers of the structure rearrangement, no matter the transformation process is exothermic or endothermic. For a given composition, glass phase has a higher internal energy than crystal.^{4, 5} Due to their energetic instability, glasses are potentially convertible to crystals.⁶ However, this conversion theoretically requires energy to accomplish and the temperature of crystallization is usually higher than the glass transition temperature.^{7, 8}

GeO_2 glass, as one of the most common oxide glasses, is an important material in the telecommunications and optic industries, due to its wider transparent band, lower transmission loss and higher refraction than SiO_2 .⁹ Since GeO_2 glass was first synthesized by Dennis and Laubengayer in 1926,¹⁰ numerous studies have been carried out in the last decades, in terms of its structure and properties.^{2, 11, 12} Simultaneously, GeO_2 nanomaterials are a potential candidate as nano-connection for the next-generation optoelectronic communications. A variety of

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3 methods, such as sol-gel reaction,¹³ thermal evaporation,¹⁴ reverse micelle process,¹⁵ chemical
4 precipitation,¹⁶ electrospinning and template methods, have been explored to synthesize GeO₂
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6 nanocrystals with different low dimensions, including nanowires,¹⁷ nanotubes,¹⁴ nanofibers¹⁸ and
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8 hollow nanostructures.¹⁹ Almost all these methods are based on the conventional dissolution-
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10 precipitation mechanism. With this mechanism, the preparation of GeO₂ nanomaterials requires
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12 basic solutions or organic germanium compounds such as Ge(OC₂H₅)₄,²⁰ since the solubility of
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14 GeO₂ in water is very low.²¹ Additionally, GeO₂ nanomaterials can also be obtained through
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16 crystallization of GeO₂ glass, but it inevitably requires high energy. The temperature for
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18 crystallization of GeO₂ glass is up to 810°C.⁷

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24 Recently, we discovered a direct transformation of GeO₂ glass to hexagonal GeO₂
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26 nanocrystals in the presence of water at room temperature and atmospheric pressure. The process
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28 of this phase transformation was found to involve a series of continuous chemical reactions
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30 without forming homogeneous solutions, which is completely different from the traditional glass
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32 crystallization through heating and classical dissolution-precipitation mechanism. It not only
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34 offers a low cost and scalable method to produce GeO₂ nanocrystals but also challenges the
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36 classic theories of glass crystallization.

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40 The evolution of GeO₂ from amorphous to crystalline structure was monitored by using X-
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42 ray diffraction (XRD). Figure 1a shows XRD patterns of the samples before and after being in
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44 contact with water for different time durations. As expected, the GeO₂ glass powder shows a
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46 broad peak (sample i), confirming its amorphous nature. Shortly after 5 min, a peak at about 26°
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48 is clearly observed (sample ii), which is diffracted from the (101) lattice plane of hexagonal
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50 GeO₂. After 15 min, all typical diffraction peaks associated with the hexagonal GeO₂ are visible
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52 from the XRD pattern (sample iii), without the presence of peaks of any impurities. With more
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time, the gradually increased intensities of the sharp diffraction peaks indicate the higher crystallinity of GeO_2 (sample iv-v). This result is in good agreement with the evolution of GeO_2 morphology as a function of time duration (Figure 1b, four sequential SEM images). After 1 h, particles with cubic shape are formed.

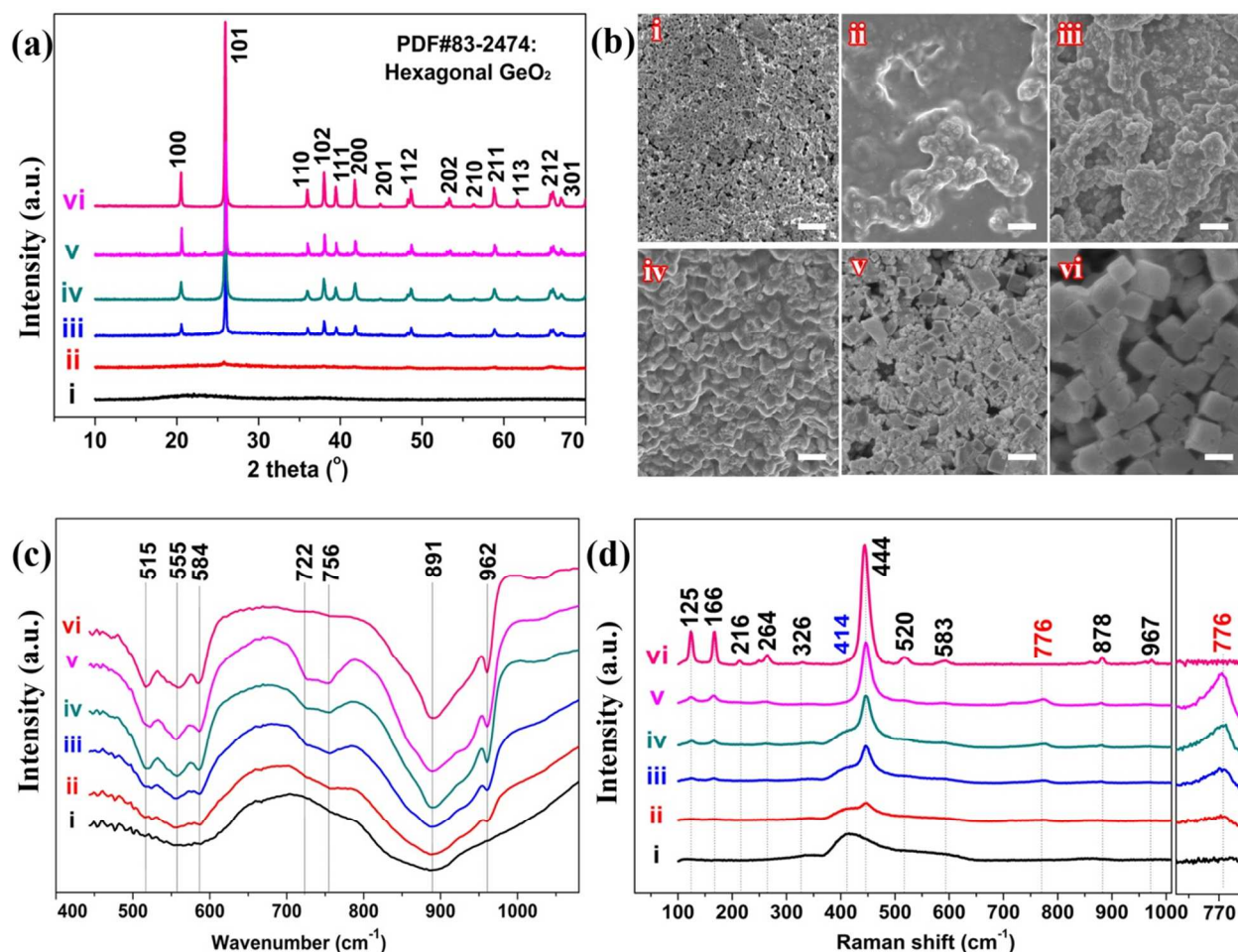


Figure 1. Structure evolution during the phase transformation of GeO_2 from amorphous state to hexagonal crystal: a) XRD patterns, b) SEM images, c) FTIR transmittance spectra and d) In-situ Raman spectra where i) the as-sieved GeO_2 glass powder, ii-v) GeO_2 glass powder with deionized water for 5 min, 15 min, 30 min and 60 min, respectively, vi) dried hexagonal GeO_2 for reference. Scale bar in the SEM images is 500 nm.

FTIR was also performed to track the phase transformation process. Figure 1c shows the FTIR spectra collected from the GeO_2 glass powder before (sample i) and after being hydrated in water for different time durations (sample ii-v), together with the standard FTIR spectrum of hexagonal GeO_2 (sample vi). The peak at 891 cm^{-1} with a shoulder at 962 cm^{-1} is associated with the ν_3 mode of $[\text{GeO}_4]$.^{20, 22} The band at 722 cm^{-1} is ascribed to hydroxyl groups²³ and the broad band in the range of $560\text{-}620\text{ cm}^{-1}$ of the glass powder (sample i) is attributed to ν_4 mode of $[\text{GeO}_4]$.^{20, 24} Sharp triplets (515 , 555 and 584 cm^{-1}) are clearly visible in (sample ii-vi), which are identified as the three absorption peaks of hexagonal GeO_2 ,^{23, 25} confirming the formation of GeO_2 with α -quartz crystal structure.

It is worth to mention that a new peak at 756 cm^{-1} is observed in the hydrated samples (sample ii-v), which has been reported to be attributed to hydroxyl group.²³ Further studies have indicated that this peak is resulted from the stretching mode of Ge-O in $[\text{GeOH}]^{3-}$, which is present in GeO_2 precursor in the form of $\text{Ge}(\text{OH})_4$, $\text{GeO}(\text{OH})_2$ or $[\text{GeO}_2(\text{OH})_2]^{2-}$.^{25, 26} This suggests that, instead of being a simple dissolving process, there must be chemical reactions occurring between the GeO_2 glass and water, prior to the formation of GeO_2 nanocrystals.

The phase transformation process was further examined by using Raman spectroscopy. Figure 1d illustrates structure evolution of the transition of GeO_2 from glass to crystal. Only one broad peak at 414 cm^{-1} is observed in the spectrum of the GeO_2 glass powder (sample i), which is attributed to the symmetric stretching mode of the Ge-O-Ge bridging oxygen.²⁷ The broad peak suggests that the Ge-O-Ge inter-tetrahedral angles of GeO_2 glass have a wide distribution, demonstrating the amorphous nature of the glass powder. Upon the addition of water to the glass powder, pronounced changes in intensity and position of the peaks are observed (sample ii-v). In comparison with the standard spectrum (sample vi), almost all the main peaks of the sample after

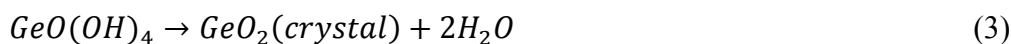
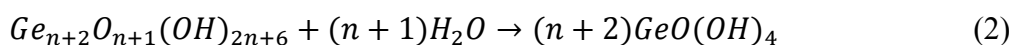
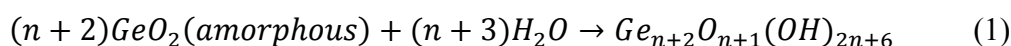
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3 reaction for 5 min can be ascribed to the hexagonal crystalline GeO_2 . The most prominent peak
4 at 444 cm^{-1} and the bands at 216 , 264 and 878 cm^{-1} are associated with the symmetric stretching
5 of Ge-O-Ge (A_1 mode).²⁸ The shifts at 326 and 583 cm^{-1} correspond to the E(TO) mode of the
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10 GeO_4 tetrahedral. The two bands at 520 and 967 cm^{-1} are attributed to the E(LO) mode, while
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12 those in the low frequency region including 125 and 166 cm^{-1} are originated from the E(LO+TO)
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14 mode.²⁸
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17 Moreover, a weak peak at 776 cm^{-1} is seen to appear in all the hydrated samples. To the best
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19 of our knowledge, this Raman peak of GeO_2 has not been reported in literatures. However, the
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21 Raman shift was detected in $\text{Sr}_2\text{ZnGe}_2\text{O}_7$,²⁹ which is assigned to the fully symmetric A_1 -vibration
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23 mode of the two $[\text{GeO}_3]$ components in $[\text{Ge}_2\text{O}_7]$ group, as further confirmed by Hanuza *et al.*³⁰
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25 and Nagabhusan *et al.*³ In our present study, multiple $[\text{GeO}_3]$ components are formed when the
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27 GeO_2 glass powder is in contact with water, due to the depolymerization of the network of GeO_2
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29 glass with a structure unit of $[\text{GeO}_4]$.
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33 The results of XRD, SEM, FTIR and Raman clearly demonstrate the phase transformation
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35 of GeO_2 glass once it is in contact with water. They also reveal that, during the transition process,
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37 intermediate substances, such as $\text{Ge}_2\text{O}(\text{OH})_6$ (or $\text{H}_6\text{Ge}_2\text{O}_7$), $\text{Ge}(\text{OH})_4$ (or H_4GeO_4) and
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39 $\text{GeO}(\text{OH})_2$ (or H_2GeO_3), are formed. It is well known that GeO_2 has three predominant
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41 polymorphs, which are amorphous, hexagonal (α -quartz structure) and tetragonal (rutile
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43 structure). The tetragonal form is practically insoluble in water, while the hexagonal and
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45 amorphous ones have very low solubility at $25\text{ }^\circ\text{C}$ (0.45% and 0.52% , respectively).^{21, 31}
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47 Although Peshko claimed that GeO_2 glass is readily soluble in water, no specific solubility data
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49 had been provided.³² Gross and Tomozawa also pointed out that pure GeO_2 glass has a strong
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51 affinity to water and thus is easily corroded when it is in contact with water.²⁴ Based on our
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experiments, it can be concluded that the high solubility of GeO_2 in water is not attributed to physical dissolution but chemical reactions. To further clarify this, an experiment was carried out by immersing a piece of 1.105 g GeO_2 glass in 20 ml water. According to the theoretical solubility of GeO_2 glass in water, the weight loss of GeO_2 is calculated to be 0.104 g. However, our result shows that the entire GeO_2 glass disappeared eventually (Figure S1), which proves that the physical dissolution of GeO_2 in water is negligible. The formation of intermediate substances during the phase transformation indicates that some chemical reactions must have been involved in the process.

The possible chemical reactions are proposed as follows:



It is well known that water weakly disassociates to H^+ and OH^- . When GeO_2 glass is in contact with water, H^+ ions approach to the oxygen atoms on the glass surface to form hydroxyl groups, whilst OH^- ions break the bridge oxygen of Ge-O-Ge to produce hydrolysis groups, $\text{Ge}_{n+2}\text{O}_{n+1}(\text{OH})_{2n+6}$, such as $\text{Ge}_4\text{O}_3(\text{OH})_{10}$, $\text{Ge}_3\text{O}_2(\text{OH})_8$ and $\text{Ge}_2\text{O}(\text{OH})_2$ (Eq. 1), which have been observed in hydrothermally grown germanates.³³ It is this process that results in the presence of the characteristic peaks at 776 cm^{-1} in the Raman spectra and 756 cm^{-1} in the FTIR spectra of the hydrated samples, as discussed above. With further hydrolysis, all the bridge oxygens in the hydrolysis groups are broken, leading to the formation of $\text{Ge}(\text{OH})_4$ (Eq. 2). Hydrolysis of GeO_2 glass results in a rapid increase in the concentration of $\text{Ge}(\text{OH})_4$. As the content of $\text{Ge}(\text{OH})_4$ reaches the critical level,³⁴ $\text{Ge}(\text{OH})_4$ starts to decompose, leading to the precipitation of hexagonal GeO_2 from the solution due to its very low solubility in water. The accumulation of

GeO₂ will result in the formation of GeO₂ crystalline nanoparticles. In addition, the water resulted from the decomposition of Ge(OH)₄ will further accelerate the hydrolysis of the amorphous GeO₂, which is like an auto-catalytic process to facilitate the continuous growth of crystalline GeO₂ nanoparticles. Therefore, water serves as a catalyst and the cycle of water ensures the constant conversion of the GeO₂ glass to nanocrystals. This is why the phase transformation of GeO₂ from glass to crystal can be accomplished when GeO₂ glass in contact with water, no matter the amount of water is small (Figure 1) or large (Figure S1). Obviously, no external energy has been involved in this phase transformation, while homogeneous aqueous solution is also not formed as required by the traditional crystallizations in aqueous environments. Therefore, this phase transformation neither follows the conventional glass crystallization process nor abides by the classical pathway of precipitation from dissolution.

Further evidence of the GeO₂ glass dissolution in water due to the chemical reactions has been collected by monitoring the change in pH value during the progress of the reactions, as shown in Figure S2. The rapid drop in pH value from 7.0 to 4.7 within 10 min is attributed to the consumption of OH⁻, due to the formation of hydrolysis groups Ge_{n+2}O_{n+1}(OH)_{2n+6}. Then, the pH value slightly decrease to 4.5 at 18 min, as a consequence of further dissociation of the hydrolysis groups into Ge(OH)₄. After that, the pH value stops decreasing when the glass powder is completely consumed. Finally, the ramping up of pH value with time is correlated to the formation of the hexagonal crystalline GeO₂.

Thermodynamic process of the phase transformation is present in Figure S3. The standard formation enthalpies of vitreous GeO₂ and hexagonal crystalline GeO₂ are -539.0 and -554.7 kJ/mol, respectively.³⁵ ΔG of the overall phase transition is -15.7 kJ/mol, indicating the transformation of GeO₂ glass to hexagonal crystalline GeO₂ is a spontaneous process. The

standard formation enthalpy of H_2O is known to be -285.8 kJ/mol and the ΔG between Ge(OH)_4 and hexagonal GeO_2 is -7.8 kJ/mol .³⁴ The ΔG of decomposition process of vitreous GeO_2 into Ge(OH)_4 in water is calculated to be -7.9 kJ/mol . Therefore, the fact that the presence of water can trigger the multiple step reactions leading to the phase transition of GeO_2 glass to crystalline GeO_2 is thermodynamically proved. Meanwhile, water provides a favorable condition to ensure rapid diffusion of the Ge-bearing species, thus promoting the phase transformation.

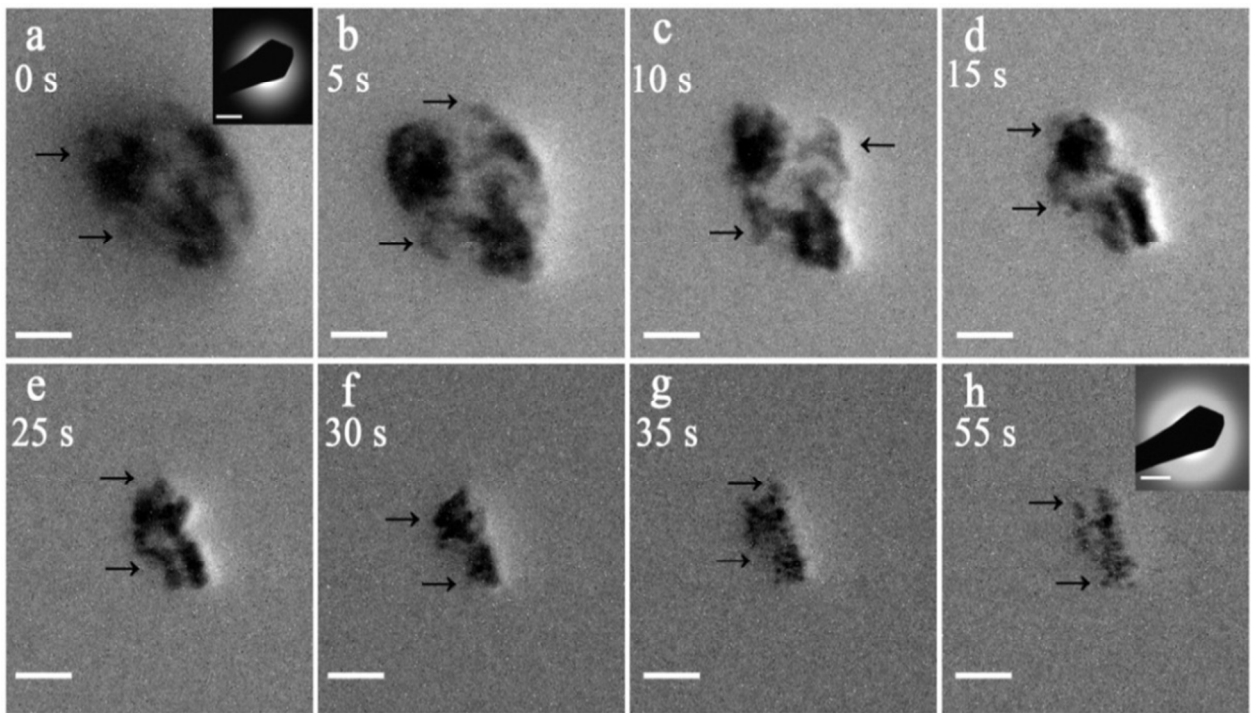


Figure 2. In-situ TEM observation of hydrolysis of the GeO_2 glass particles in deionized water. a) Original glass particles in water. b-h) Sequential in-situ TEM images after reaction for different times in water. Amorphous nature of the GeO_2 glass particles is confirmed by the corresponding diffraction pattern (inset of a) and h)). Scale bars are 100 nm in a-h) and 2 nm^{-1} in the insets of a) and h).

To gain further evidence to support the above explanation, we use in-situ transmission electron microscopy (TEM) to directly observe the microstructural evolution of GeO_2 during the phase transformation process. The in-situ TEM with a liquid cell holder is shown schematically in Figure S4. Figure 2 is a sequence of TEM images of the GeO_2 glass particles in deionized water for different times, revealing the hydrolysis process. It is clearly demonstrated that the GeO_2 glass particles intensively react with water. As a result, size of the particles is quickly reduced from ~ 200 nm to ~ 50 nm after about 0.5 min (Figure S5 and Video S1). Amorphous nature of the particles is confirmed by the diffraction patterns (insets of Figure 2a, 2h). Hydrolysis of GeO_2 glass leads to the production of a large number of tiny nanodots which surround the GeO_2 particles. These nanodots become more and more visible (Figure 3 and Figure S6) after the GeO_2 glass nanoparticles are sufficiently hydrolyzed in water. The nanodots have an average size of ~ 0.5 nm (Figure S7). Close accumulation of these nanodots forms a spherical dense liquid cluster (DLC) with a size of about 150 nm.

It is of particular interest that these nanodots act as a precursor for the nucleation of GeO_2 crystals. As an example, Figure 3 shows seven bright-field TEM images taken at 0 s, 5 s, 15 s, 25 s, 35 s, 270 s and 420 s, respectively, which are retrieved from the original images, as shown in Figure S6. This is a direct evidence of the crystallization of GeO_2 , which involves nucleation and growth steps. This continuous process is clearly demonstrated in Video S2. Figure 3 also reveals that the nanodots cannot directly serve as nuclei for the growth of GeO_2 crystals, since they are much smaller than the critical nucleus radius and thus cannot meet the thermodynamic and kinetic conditions for homogeneous nucleation. This is confirmed by the presence of the spherical DLC-1 in Figure S6, which becomes gradually bright in the TEM image with time and is almost invisible after water flow for 270 s. Obviously, crystallization is not started. In contrast,

the DLC-2 is fully transformed to nanocrystals within 270 s. Different from DLC-1, the DLC-2 still contains several fairly large particles (such as 1, 2 and 3), which are original GeO_2 glass and have not been completely hydrolyzed. These particles remain sufficiently high surface energy and thus attract the nanodots nearby, which are stacked orderly with a specific lattice orientation, as evidenced by the diffract pattern (Figure 3h). The particles are identified to be hexagonal GeO_2 according to the index of the diffraction pattern, in agreement with the XRD results.

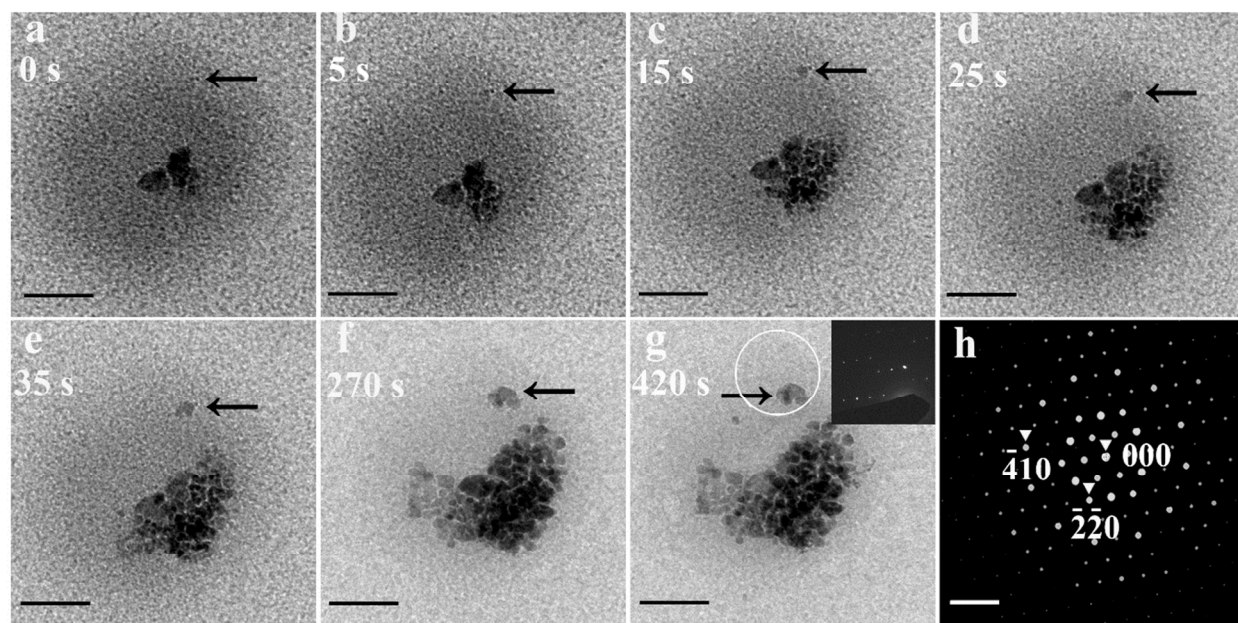


Figure 3. In-situ TEM observation of nucleation and crystallization of the hexagonal GeO_2 nanocrystals in deionized water. a) A dense liquid cluster with a size of about 150 nm developed around several glass particles that have not been completely dissolved. b-g) Sequential in-situ TEM images of particle hydrolysis, accompanied by the nucleation and crystallization for different reaction times in water. The inset in g) is a selected-area diffraction pattern of the crystal. h) Simulated and indexed diffraction patterns of hexagonal GeO_2 . The images of a-f) are retrieved from Figure S7. Scale bars are 50 nm in a-g) and 5 nm⁻¹ in h).

Based on the chemical reactions discussed above, it can be concluded that the presence of the clusters comprising of $\text{Ge}(\text{OH})_4$ is the onset of crystallization. The aggregation of these clusters leads to the formation of DLCs. Recently, a two-step crystallization mechanism was proposed, in which the crystallization is fulfilled due to the aggregation of clusters³⁶⁻³⁸ instead of the assemble of atoms or molecules.³⁹ In this mechanism, the DLCs are regarded as necessary locations for crystal nucleation.⁴⁰ However, so far, the dynamic process of nucleation from solutions, due to aggregation of clusters at the sites of DLCs, has not been directly observed. In the present study, our results show that the continuous formation and the growth of nuclei in the DLC rapidly consume the clusters until they become exhausted (Figure 3f, g). Thus, it is clear now that the clusters behave as the building units for nucleation, whilst the DLCs provide favorable yet not-inevitable prenucleation sites for the formation of nuclei (such as DLCs-1, 2 in Figure S6). When solid-liquid interfaces are preserved in the DLCs, nuclei can be formed preferentially there. Otherwise, a critical concentration of clusters in the DLCs is required to implement the nucleation.^{41, 42}

Figure 4a shows an in-plane view schematic diagram that describes the detailed formation process of the GeO_2 nanocrystals as a result of the reaction between the GeO_2 glass and water. The planar structure model of amorphous and crystalline GeO_2 is established based on the similar structure of SiO_2 .⁴³ Once a GeO_2 glass particle is in contact with water, the particle surface is quickly hydrolyzed into anionic groups by breaking the bridge oxygen of Ge-O-Ge (Figure 4a(i), a(ii)), leading to stoichiometric dissolution⁴⁴ of the GeO_2 glass. Oscillatory spatial patterns of the new surface of the partly hydrolyzed GeO_2 glass block, resulted from atomically sharp reaction interface,⁴⁵ can be clearly observed (Figure S8). The newly formed surface continuously reacts with water, so that the bridge oxygens in the anionic groups are continuously

broken, until the entire particle is hydrolyzed to $\text{Ge}(\text{OH})_4$ (Figure 4a(iii)). However, the stability of $\text{Ge}(\text{OH})_4$ is significantly influenced by pH value of the solution.³⁴ The $\text{Ge}(\text{OH})_4$ solution is stable and transparent under basic environment (Figure S9-b), while the acidic conditions are favorable for the aggregation of $\text{Ge}(\text{OH})_4$ into multimers forming an opaque white suspension (Figure S9-c).

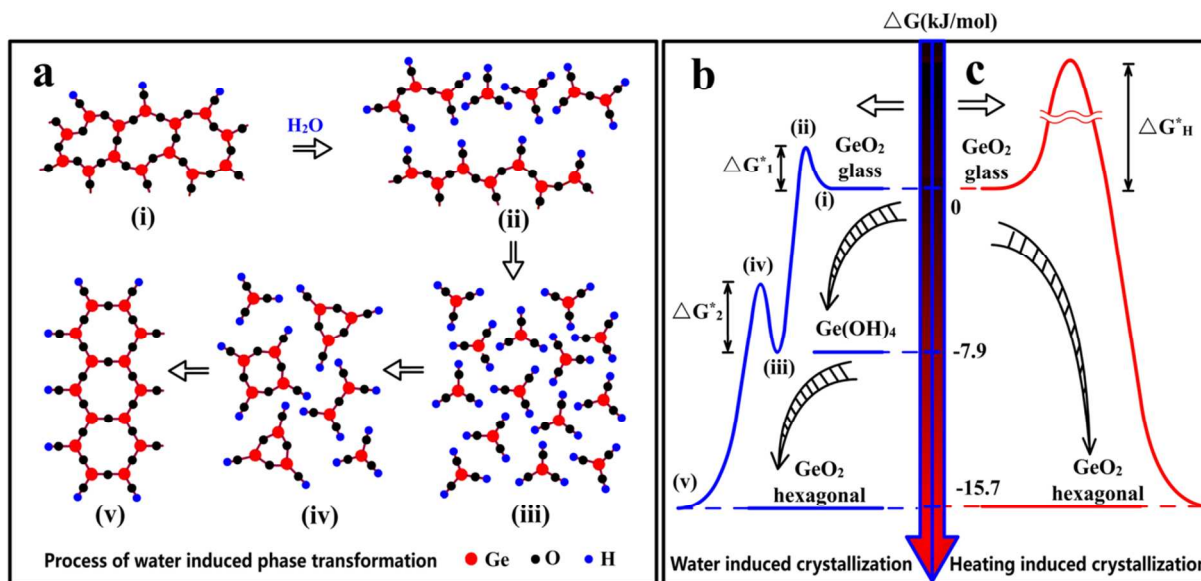


Figure 4. Atomic arrangement evolution process of the water induced GeO_2 glass phase transformation and the comparative kinetic analysis of phase transformations induced by water and heating. a) In-plane diagram showing the process of water induced GeO_2 glass phase transformation. b) The corresponding kinetic process of water induced transformation. c) Kinetic analysis of heating induced crystallization of GeO_2 glass. GeO_2 glass network structure with a given surface profile ((i) in a), decomposition of the network due to the attack of its surface by OH^- ions in water, forming a large variety of anionic groups and generating new surface ((ii) in a), further decomposition of the anionic groups into $\text{Ge}(\text{OH})_4$ ((iii) in a), formation of clusters with possible structures consisting of dimers, 3-rings and double 4-rings (D4R) due to the aggregation of $\text{Ge}(\text{OH})_4$ ((iv) in a), and the presence of prenucleation clusters or nucleus owing

to the further aggregation of the clusters ((v) in a). All atoms of Ge and O are arranged in GeO_3 triangles corresponding to GeO_4 tetrahedra in 3D. ΔG_1^* , ΔG_2^* and ΔG_H^* are the energy barriers for breaking the bridge oxygen of Ge-O-Ge, water removal step of the clusters formation and the nucleation by heating, respectively.

Germanium, in its tetrahedral coordination geometry with oxygen, forms longer bonds (~ 1.74 Å) with smaller bond angles (Ge-O-Ge angle $> 130^\circ$), as compared with the analogous silicon oxide geometries (~ 1.61 Å and $> 140^\circ$), leading to the formation of 3-ring and D4R as specific building units in germanates.⁴⁶ Based on the size and distribution of the cluster in the frames from Video S2 (Figure S7), their half size is in the range of 0.4 - 0.6 nm, which are close to the edge length of the 3-ring⁴⁶ and D4R.³⁴ Therefore, the clusters are most likely to be 3-ring ($\text{Ge}_3\text{O}_3(\text{OH})_6$) and D4R ($\text{Ge}_8\text{O}_{12}(\text{OH})_8$) (Figure 4a(iv)). The dimer ($\text{Ge}_2\text{O}(\text{OH})_6$), a transitional structure from monomer to ring structure, is also possibly present. Theoretical studies have shown that the formation of the dimer of $\text{Ge}(\text{OH})_4$ is an initial step during the production of germanates in solutions.⁴⁷ Since the energy to overcome the barrier of water removal depends only on the weak hydrogen bonding network surrounding the dimer,⁴⁷ it is much lower than the energy required by the traditional crystallization through heating. Due to the aggregation and water removal of 3-ring or D4R, the clusters continue to grow into nuclei (Figure 4a(v)). The relative number ratio of the clusters at each size is remained nearly the same throughout the clusters aggregation (Figure S7). However, it is necessary to mention that the number of clusters with the size of 1.1 nm is larger than those of the ones having the size of over 0.8 nm. The possible reason for the presence of these unexpected 1.1 nm clusters is that they are stable and can survive in the solution, thus acting as prenucleation clusters.

All the results show that the presence of water provides favorable kinetic conditions for the transformation of GeO₂ from amorphous to crystalline phase. Due to the difficulty in diffusion of atoms in solid matters, a temperature up to 810°C is required to overcome the huge energy barrier (ΔG_H^* , 276 kJ/mol) for nucleation of GeO₂ glass⁷ (Figure 4c). In sharp contrast, our present study demonstrates that the energy barrier has to be overcome in the phase transformation of GeO₂ glass induced by water is much lower. This is because both the activation energies to break the Ge-O-Ge bond (ΔG_1^*) and to remove water from the multiple Ge(OH)₄ monomers (ΔG_2^*) are very low, since they can be satisfied just by the hydrogen bonding network surrounding the Ge-bearing species (Figure 4b).⁴⁷

In summary, GeO₂ glass is readily transformed to nanocrystals in the presence of water at ambient conditions, which has been clearly demonstrated by using XRD, SEM, FTIR and Raman spectroscopy techniques and further confirmed with the direct observation by using in-situ liquid-cell TEM. A series of continuous chemical reactions between the amorphous GeO₂ and water are clarified to be responsible for the transformation of GeO₂ glass, involving the hydrolysis of amorphous structure to develop dense liquid clusters, which act as a precursor of the subsequent nucleation and growth of GeO₂ crystalline nanoparticles. This is a new phase transformation process of oxide glass that provides a significant complementation to the classic crystallization theories. It also brings out an efficient and cost-effective route to manufacture GeO₂ nanocrystals and hopefully applicable to other nanomaterials.

ASSOCIATED CONTENT

Supporting Information

The supporting information includes sample preparation, characterization, 9 figures and 2 videos.

AUTHOR INFORMATION

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AUTHOR CONTRIBUTIONS

Z. X. conceived the study and wrote the paper. Y. H. and L. B. K. designed the experiments and revised the paper. Z. X carried out the data analyses with input and advice from X. S. and Y. W. The characterization experiments were conducted by X. L. and X. L. L. The in-situ TEM experiments were conducted by Z. W. with the assistance of Z. S. The indexing of diffraction pattern was completed by B. Z. All authors contributed to discussion of the results.

NOTES

The authors declare no competing financial interest.

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